

PROJECT INTRODUCTION

Project Title: Smart Route Finder

- One-line Idea: Intelligent route engine that finds and explains optimal travel paths.
- Problem Domain: Route Optimization & Graph Algorithms
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THE CHALLENGE

- Real-world problem: Finding the best route among multiple alternatives.
- ■ Why it matters: Saves time, cost, and improves decision-making.
- ■ Challenges: Multiple routes, varying costs, scalability, explainability.

OUR INNOVATION

- Compares multiple routes instead of showing only one.
- ■ Explains why the selected route is optimal.
- ■ Key Features: Alternative routes, cost comparison, visual path explanation.
- ■ Uses Dijkstra, Floyd-Warshall, and A* algorithms.

HOW IT WORKS

- Input: Locations, distances, travel costs
- ■ Processing: Build graph representation
- ■ Algorithms: Dijkstra / Floyd-Warshall / A*
- ■ Output: Optimal route, alternative routes, explanation
- ■ User Journey: Select source → destination → view results

TECHNOLOGY & ALGORITHM

- Tech Stack: Python/Flask, React, Leaflet, SQLite
- ■ Algorithms: Dijkstra, Floyd-Warshall, A*
- ■ Why Chosen: Efficiency, shortest-path computation, heuristic search
- ■ Complexity: Dijkstra $O((V+E)\log V)$, Floyd-Warshall $O(V^3)$, A* depends on heuristic

DEMO & IMPACT

- Demo: Route comparison dashboard
- ■ Sample Output: Best path with total cost and alternatives
- ■ Performance: Fast route computation and visualization
- ■ Benefits: Transparency, better planning, reduced travel cost

FUTURE VISION

- Current Achievement: Working algorithm-centric prototype
- ■ Future Improvements: Real-time traffic, GPS integration, ML predictions
- ■ Scalability: Support larger maps and datasets
- ■ Final Takeaway: Explainable and optimized route planning
- ■ Thank You / Q&A